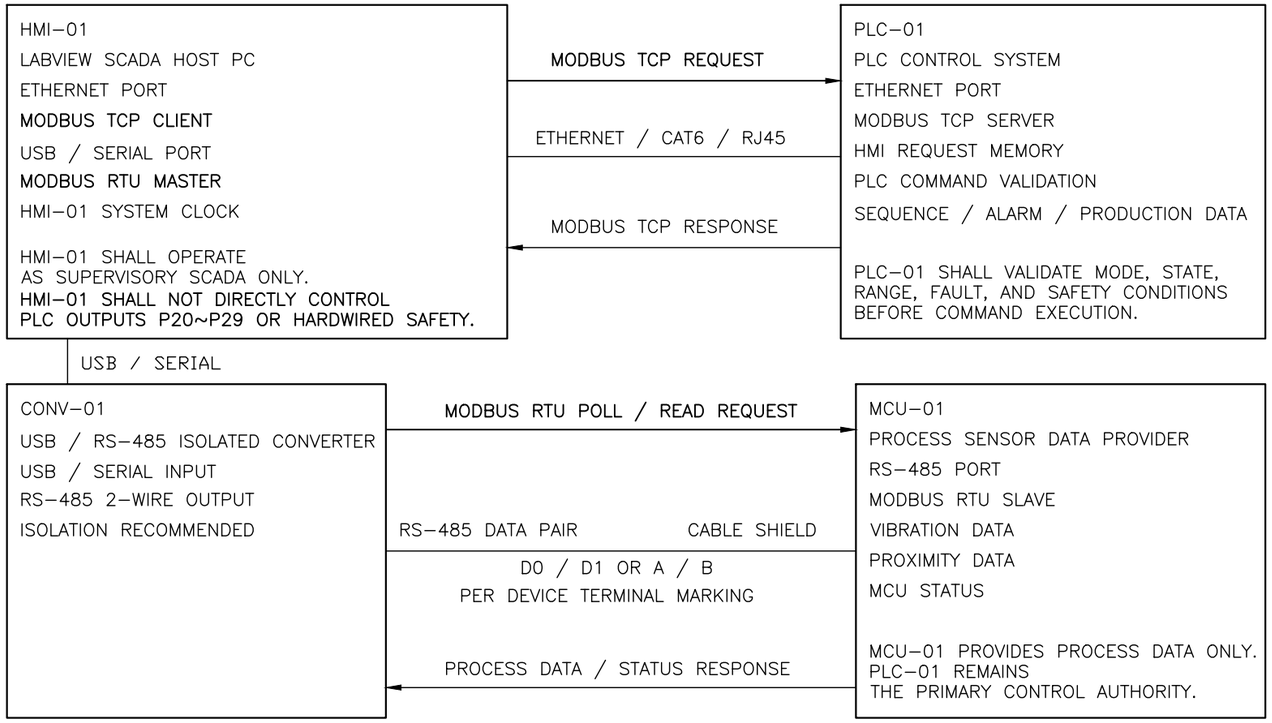
HMI–01 / PLC–01 / MCU–01 COMMUNICATION NETWORK MODBUS TCP / ETHERNET AND MODBUS RTU / RS–485 INTERFACE

A. SYSTEM COMMUNICATION TOPOLOGY



RS–485 DATA PAIR POLARITY SHALL MATCH THE ACTUAL TERMINAL MARKING OF CONV–01 AND MCU–01.

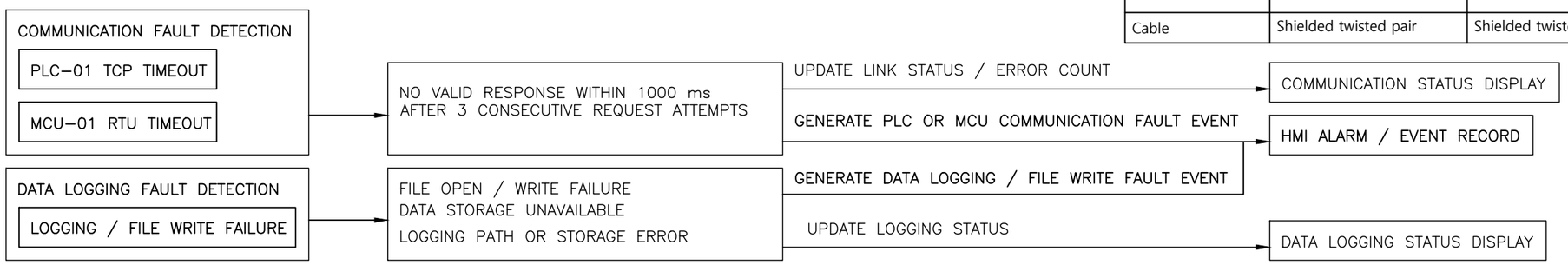
B. MODBUS TCP / ETHERNET INTERFACE

MODBUS TCP / ETHERNET INTERFACE			
Parameter	HMI-01	PLC-01	Design value
Network role	Modbus TCP Client	Modbus TCP Server	HMI initiates all transactions
IPv4 address	192.168.10.10	192.168.10.20	Static addressing
Subnet mask	255.255.255.0	255.255.255.0	Dedicated local subnet
Default gateway	N/A	N/A	No external routing required
TCP port	Client connection	502	Standard Modbus TCP port
Unit identifier	1	1	Project configuration value
Poll interval	250 ms nominal	Response required	Per 018 Sheet 2
Response timeout	1000 ms	-	Communication fault threshold
Retry count	3 attempts	-	Before HMI communication fault
Cable / connector	CAT6 / RJ45	CAT6 / RJ45	Shielded industrial cable recommended

C. MODBUS RTU / RS–485 INTERFACE

MODBUS RTU / RS-485 INTERFACE			
Parameter	HMI-01 / CONV-01	MCU-01	Design value
Network role	Modbus RTU Master	Modbus RTU Slave	HMI initiates polling
Physical layer	USB to RS-485 converter	RS-485 2-wire	Half-duplex differential link
Slave address	Master	1	MCU-01 Modbus slave ID
Baud rate	9600 bps	9600 bps	Must match both ends
Frame format	8 data / no parity / 1 stop	8N1	Must match both ends
Poll interval	250 ms nominal	Response required	Per 018 Sheet 2
Response timeout	1000 ms	-	Fault threshold
Retry count	3 attempts	-	Before HMI alarm event
Termination	120 Ohm at end of bus	120 Ohm at end of bus	120 Ohm Across data pair At two physical end points only
Cable	Shielded twisted pair	Shielded twisted pair	Data pair + Cable shield

D. COMMUNICATION HEALTH / FAULT BOUNDARY



COMMUNICATION FAULT SHALL CLEAR ONLY AFTER VALID COMMUNICATION RESPONSE IS RESTORED.

DATA LOGGING FAULT SHALL CLEAR ONLY AFTER SUCCESSFUL FILE WRITE OR STORAGE ACCESS IS RESTORED.

COMMUNICATION FAULT SHALL NOT BYPASS, DISABLE, OR REPLACE HARDWIRED SAFETY FUNCTION.

[INSTALLATION / SAFETY NOTES]

1. HMI–01 SHALL BE THE MODBUS TCP CLIENT FOR PLC–01 AND MODBUS RTU MASTER FOR MCU–01.
2. PLC–01 SHALL VALIDATE ALL HMI REQUESTS BEFORE COMMAND EXECUTION.
3. HMI–01 SHALL NOT DIRECTLY WRITE PLC OUTPUTS P20~P29.
4. HMI COMMUNICATION SHALL NOT REPLACE, OVERRIDE, OR BYPASS HARDWIRED SAFETY FUNCTION.
5. RS–485 A/B POLARITY, BAUD RATE, FRAME FORMAT, SLAVE ID, AND TERMINATION SHALL MATCH BOTH END POINTS.
6. STATIC IP ADDRESS VALUES SHALL BE DOCUMENTED AND MAINTAINED AS PROJECT CONFIGURATION DATA.
7. ALL MODBUS REGISTER / BIT / DATA TAG DEFINITIONS: SEE DWG 015 / SHEET 1 OF 3, SHEET 2 OF 3, AND SHEET 3 OF 3.
8. HMI FUNCTIONAL DATA FLOW / ALARM / LOGGING ARCHITECTURE: SEE DWG 017.
9. CABLE SHIELD SHALL NOT BE USED AS RS–485 SIGNAL COMMON.
   CONNECT SIGNAL COMMON / 0V ONLY IF REQUIRED BY THE ACTUAL CONV–01 AND MCU–01 DEVICE MANUALS.
10. CONNECT CABLE SHIELD TO FUNCTIONAL EARTH AT ONE END ONLY, UNLESS DEVICE INSTALLATION MANUAL SPECIFIES ANOTHER APPROVED METHOD.
11. POLL INTERVAL, TIMEOUT, AND RETRY VALUES SHALL BE VERIFIED DURING COMMISSIONING.

[INTERFACE LEGEND]
- CYAN TEXT
- DRAWING TITLE / CROSS REFERENCE / TAG REFERENCE
- SOLID LINE
- PHYSICAL COMMUNICATION LINK
- ARROWHEAD
- DATA / COMMAND FLOW DIRECTION

[INTERFACE LEGEND]
- YELLOW ARROW
- COMMAND / REQUEST / POLL DIRECTION
- GREEN ARROW
- STATUS / RESPONSE / PROCESS DATA
- RED TEXT / ARROW
- FAULT / ALARM / SAFETY RELATED CONDITION

NO.	DATE	REVISION
A	2026–09–22	INITIAL RELEASE

DWG TITLE

EDS Wafer Map Linked Package
Final Test and Multi–Bin Sorting Cell

SCADA COMMUNICATION NETWORK / PHYSICAL INTERFACE DIAGRAM

DRAWN BY	DATE
J. H. KO	2026–09–22

DWG NO: 018

SHEET NO: 1 OF 2